

GRCF: Group-wise Ranking and Calibration Framework for Multimodal Sentiment Analysis

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Abstract

Most multimodal sentiment analysis algorithms focus on point-wise regression. While straightforward, it uses absolute values to represent subjective sentiment and overlooks the ordinal nature of sentiment, resulting in unstable predictions and poor correlation. Inspired by the way humans perceive sentiment, pair-wise ordinal learning frameworks emerged to address this gap, capturing relative order by learning from comparisons. However, they have two limitations: First, they assign uniform importance to all pairs, failing to adaptively focus on hard-to-rank samples. Second, they employ static ranking margins, which fail to reflect the varying semantic distances between sentiment groups. To address this, we propose a Group-wise Ranking and Calibration Framework (GRCF) that introduces adaptive error-weighting and dynamic margin mechanism, which resolves these limitations by simultaneously preserving relative ordinal structure and ensuring absolute score calibration. Specifically, Stage 1 introduces a group-aware ranking loss to build a fine-grained ordinal structure by assigning higher optimization priority to confusable pairs. Stage 2 then employs an MAE-driven objective to align the prediction magnitudes. To validate its generalizability, we extend GRCF to classification tasks, including multimodal humor and sarcasm detection. It achieves state-of-the-art performance on regression datasets while also showing strong generalizability in classification tasks.

CCS Concepts

• Computing methodologies → Scene understanding; Ranking.

Keywords

affective computing, multimodal analysis, ordinal learning

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1 Introduction

Understanding human affect through multimodal signals (language, vision, and acoustics) has been a central pursuit in affective computing. Unlike categorical perception, human emotion unfolds along a continuous, ordinal spectrum of intensity [34, 43]. Sentiment intensity is inherently comparative: terms like “desperate” are naturally

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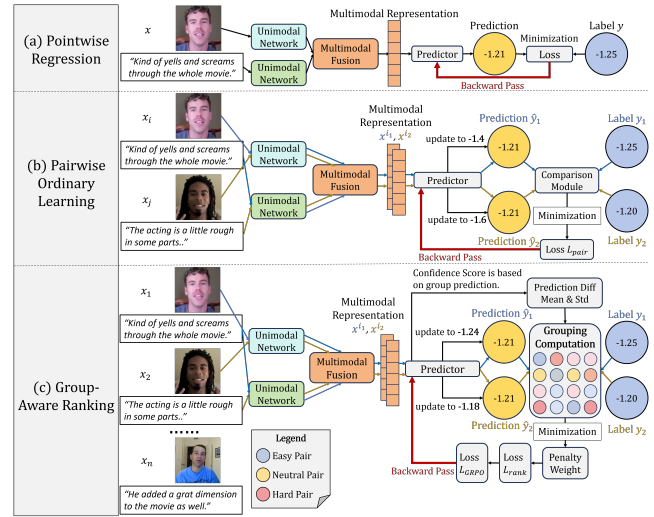


Figure 1: Comparison of optimization strategies in MSA.

understood as more negative than “sad”, yet their exact absolute magnitudes remain subjective and ambiguous.

Despite the progress achieved by multimodal transformers [4, 36, 37] and pre-trained encoders such as BERT [5], DeBERTa [12] and ALBERT [15], traditional Multimodal Sentiment Analysis (MSA) [47] frameworks predominantly rely on point-wise regression. As shown in Figure 1, these methods [6, 8, 13, 17] compute the loss between continuous labels and predicted scores, treating each instance independently. However, forcefully assigning absolute cardinal values to subjective affective concepts is neither noise-resistant nor aligned with human perception [43]. Because such independent objectives completely neglect relational ordering, point-wise models inherently struggle to capture the ordinal hierarchy of human emotions, often resulting in unstable predictions near neutral boundaries.

To overcome the limitations of point-wise regression, recent works have pivoted towards ordinal learning, typically implemented via pair-wise constraints. By evaluating pairs of samples rather than isolated instances, these models are encouraged to preserve the monotonic relative relationships between sentiments (e.g., ensuring sample i is ranked higher than sample j). Pioneering works like Trustworthy Multimodal Sentiment Ordinal Network (TMSON) [42] and Multimodal Ordinal Affective Computing (MOAC) [25] advance MSA by incorporating ordinal reasoning. However, these pair-wise methods introduce two critical limitations: First, they rely on uniform weighting, treating all pair-wise relationships equally. This allows easily distinguishable pairs to dominate the loss, overwhelming the nuanced signals from ambiguous, hard-to-rank samples. Second, they employ static ordinal constraints (margins), which fail to reflect the varying semantic distances between sentiment

groups. For example, the semantic gap between slightly positive (+0.5) and strongly positive (+3.0) is treated indistinctly from the gap between neutral (0) and slightly positive (+0.5). This crucial semantic distance information remains underexploited [29].

Inspired by how humans naturally prioritize ambiguous emotions and gauge semantic distances, we propose a two-stage Group-wise Ranking and Calibration Framework (GRCF). Our framework directly resolves the aforementioned limitations by simultaneously preserving relative ordinal structure, adapting to sample difficulty, and ensuring absolute score calibration.

In Stage 1, GRCF establishes a robust structural foundation via a Group-Aware Ranking Loss, which explicitly targets the two limitations of prior ordinal methods. First, to address the issue of uniform weighting, we introduce the Adaptive Error-Weighting mechanism. While conceptually inspired by the advantage-weighting philosophy of Group Relative Policy Optimization (GRPO) [27, 33], our approach diverges from standard GRPO, which evaluates absolute individual rewards against a group mean. Instead, we adapt the core group-wise concept into a continuous, pair-wise hard-example mining strategy. Rather than simply assessing whether sample i is greater than sample j , GRCF evaluates their pair-wise prediction gap against batch statistics, transforming this relative prediction error into an adaptive penalty weight. This mechanism forces the optimization process to prioritize hard, confusable pairs, effectively filtering out gradient noise from easy pairs. Second, to resolve the constraints of static margins, the Dynamic Margin mechanism utilizes deliberately constructed overlapping ordinal groups to model semantic ambiguity and homogeneity near decision boundaries. By adaptively computing the margin based on the group memberships of sample pairs, GRCF ensures that the optimization standard scales proportionally with the true semantic gap.

In Stage 2, a multi-objective calibration process refines the learned manifold. Building upon the robust relative ordinal structure established in Stage 1, this stage anchors the absolute prediction magnitudes to the annotated sentiment scales. By employing differential learning rates, we ensure GRCF achieves precise value alignment while preserving the previously learned pair-wise relationships.

Experiments on CMU-MOSI [47], CMU-MOSEI [48], and CH-SIMS v2 [20] demonstrate that GRCF achieves state-of-the-art (SOTA) results. Beyond regression tasks, GRCF also shows strong generalizability when extended to binary classification tasks (e.g., Multimodal Humor Detection, MHD, and Multimodal Sarcasm Detection, MSD). Our main contributions are summarized as follows:

- We devise the Error-Weighting mechanism, which controls the model to focus more on confusable samples.
- We adopt a Dynamic Margin realized through overlapping intervals, capturing both the semantic ambiguity and homogeneity near decision boundaries, and ensuring the margin scales proportionally with the true semantic gap.

2 Related Work

2.1 Point-wise MSA methods

Numerous point-wise MSA methods aim to infer human affective states by integrating textual, acoustic, and visual signals. Early fusion-based methods [13, 18, 28, 35, 49] such as Tensor Fusion Network (TFN) [45] and Memory Fusion Network (MFN) [46] explored

tensor interactions and memory mechanisms to model temporal dependencies. Transformer-based models [16, 31] like Memory Fusion Network (MulT) [36] later introduced cross-modal attention to enable fine-grained contextual alignment. With the rise of pretrained language models, recent approaches leverage pre-trained text encoders such as BERT [5], DeBERTa [12] and ALBERT [15], coupled with lightweight acoustic and visual branches. Many other models are also devised based on pretrained language model, e.g., Decoupled Multimodal Distillation (DMD) [17], Embracing Aleatoric Uncertainty (EAU)[8] and Modality-Specific Enhanced Dynamic Emotion Experts (EMOE) [6].

Despite progress, existing point-wise models fail to capture the ordinal nature of human sentiment analysis. In contrast to those point-wise models dealing with samples separately, our GRCF adopts ordinal learning to simulate the ordinal nature of human.

2.2 Ordinal and Pair-wise Ranking Learning

Sentiment prediction can be naturally formulated as an ordinal regression problem, where preserving sample order provides stronger supervision than discrete classification. Early pair-wise ranking models such as RankNet [2] and LambdaRank [32] learn monotonic score differences to maintain relative ordering. Xie *et al.* [42] devised TMSON, proposing a novel multimodal fusion framework that combines modality-specific uncertainty modeling with ordinal regression to achieve more reliable sentiment analysis in the ordinal sentiment space. Recently, Mai *et al.* [25] introduced MOAC, which formulates MSA as an ordinal learning problem by performing label-level and feature-level comparisons. Their approach demonstrates that explicitly modeling ordinal relations improves the interpretability and stability of multimodal sentiment models.

However, these pair-wise methods treat every pair equally, which contradicts human cognition. Humans tend to focus more on emotions that are harder to distinguish, and GRCF advances this direction by introducing a group-wise Error-Weighting mechanism that adaptively adjusts pair-wise constraints based on inter-group semantic distance.

3 Methodology

3.1 Multimodal Encoder Architecture

(a) *Unimodal Encoders*: Each utterance's text T is processed by a text encoder to produce a global textual representation $\mathbf{z}_t$ (using the [CLS] token output). For the audio A and video V modalities, $\mathbf{z}_a$ and $\mathbf{z}_v$ are produced via attention pooling:

$$\mathbf{z}_a = \sum_{i=1}^{L_a} \alpha_i^{(a)} \mathbf{A}_i, \quad \alpha_i^{(a)} = \frac{\exp(\mathbf{w}_a^\top \mathbf{A}_i)}{\sum_{j=1}^{L_a} \exp(\mathbf{w}_a^\top \mathbf{A}_j)}, \quad (1)$$

and analogously for $\mathbf{z}_v$.

(b) *Multimodal Fusion*: The goal is to produce an intermediate representation, $\mathbf{z}_{\text{intermediate}}$, which is then fed into a shared unified encoder. For continuous sentiment, a holistic understanding of the context and intensity from all modalities is paramount. Regression tasks, which aim to produce a single, continuous-like score, inherently require the integration of all modalities into one unified judgment, rather than modeling discrete conflicts between them. Consequently, for this type of holistic integration task, a simple and

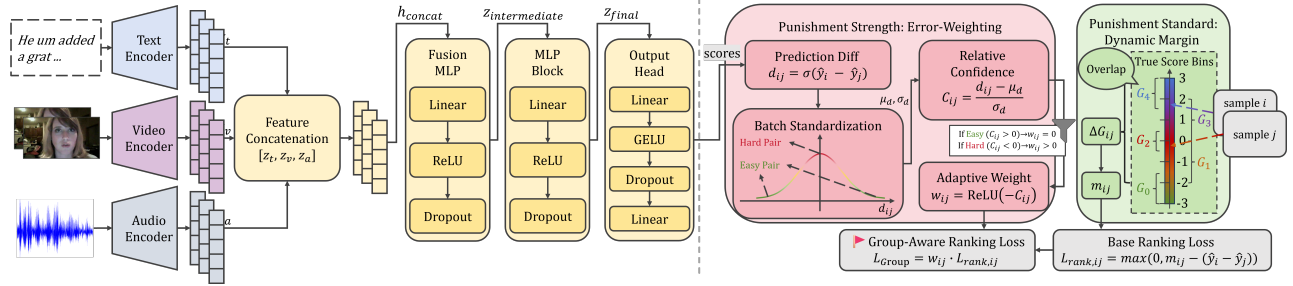


Figure 2: The forward pass of GRFC and core loss calculation which decouples the punishment standard from the strength.

efficient “Concatenate-then-MLP” architecture is highly effective. First, the unimodal representations are concatenated:

$$\mathbf{h}_{\text{concat}} = [\mathbf{z}_t; \mathbf{z}_a; \mathbf{z}_v]. \quad (2)$$

This combined vector is then passed through a dedicated fusion MLP to create the intermediate representation:

$$\mathbf{z}_{\text{intermediate}} = \text{MLP}_{\text{fuse}}(\mathbf{h}_{\text{concat}}). \quad (3)$$

$\mathbf{z}_{\text{intermediate}}$ captures the salient multimodal information. It is passed through a Unified Encoder (a MLP block) to create the definitive representation $\mathbf{z}_{\text{final}}$, which is then fed into the regression head. To predict a continuous score $\hat{y}$, the head (a two-layer MLP) constrains the output using a scaling factor α (e.g., 3.5) and a hyperbolic tangent function to match the annotation range (e.g., [-3, 3]):

$$\hat{y} = \alpha \times \tanh(f_{\text{reg}}(\mathbf{z}_{\text{final}})). \quad (4)$$

3.2 Stage 1: Structural Foundation

The goal of Stage 1 is to address the problem that traditional regression neglects relational ordering. Prior ordinal frameworks like MOAC [25] attempt to solve this with a complex approach, designing separate, explicit losses for label-level ranking ($\mathcal{L}_{\text{lo}}$), feature-level difference ($\mathcal{L}_{\text{fd}}$), and neutral-point calibration ($\mathcal{L}_{\text{neu}}$).

Instead of designing multiple, complex feature-level losses, our GRFC framework applies a single, unified loss ($\mathcal{L}_{\text{group}}$) at the final output layer. Our hypothesis is that by using an Adaptive Error-Weighting strategy (acting as continuous hard-example mining) to implicitly focus the optimization on hard-to-rank pairs (the “Punishment Strength”), the model is automatically forced to learn a robust and well-separated feature manifold. This is guided by our Dynamic Margin (the “Punishment Standard”), which directly solves the static constraint problem. To achieve this, the Stage 1 loss function combines three key components:

(a) *Group-Aware Ranking Loss ($\mathcal{L}_{\text{group}}$):* This is the core loss for modeling ordinality. Its design is a multi-step process:

Defining Overlapping Ordinal Groups ($g(\cdot)$): To enforce the “semantically proportional” structure mentioned above, we must first quantify semantic distance. Prior methods [19, 25, 38] treating all pairs with a static margin fail to do this. Our solution is to first realistically model the fuzzy and continuous nature of human sentiment by mapping a continuous score s_i to a set of ordinal groups, $g(s_i)$. We define $K = 5$ groups using overlapping intervals, which is a deliberate choice. This design allows a single sample

to belong to multiple groups, which serves two key purposes: it captures the inherent ambiguity of a single score on a boundary (for instance, a score of $s_i = 0.2$ is assigned to both G_2 and G_3), while also ensuring that numerically-close boundary samples (e.g., -1.8 and -1.6) share a group membership, correctly identifying them as semantically similar. Specifically, with scores ranging from -3 to 3, the groups are defined as: G_0 [-3.0, -1.5], G_1 [-2.0, 0.0], G_2 [-0.5, 0.5], G_3 [0.0, 2.0], and G_4 [1.5, 3.0].

Calculating the Dynamic Margin: For a pair-wise samples (i, j) with scores $s_i > s_j$, the group difference ΔG_{ij} is defined as the maximum semantic distance between their groups:

$$\Delta G_{ij} = \max_{G_a \in G(s_i), G_b \in G(s_j)} |G_a - G_b|. \quad (5)$$

Consequently, a comparison between an ambiguous sample (e.g., $s_i = 0.2$ in G_2, G_3) and a stable sample (e.g., $s_j = 2.5$ in G_4) is robustly constrained by the largest possible semantic gap (e.g., $|4 - 2| = 2$). If two samples share any group (e.g., $s_i = -1.8$ and $s_j = -1.6$, which are both in G_0 and G_1), their ΔG_{ij} becomes 0. The Dynamic Margin m_{ij} is then defined based on this group difference ΔG_{ij} :

$$m_{ij} = \begin{cases} m_{\text{intra}}, & \Delta G_{ij} = 0, \\ m_{\text{base}} + \Delta G_{ij} m_{\text{step}}, & \text{otherwise,} \end{cases} \quad (6)$$

where m_{intra} acts as a small base margin for samples within the same group to maintain a minimal local separation, m_{base} is the foundational margin for inter-group pairs, and m_{step} is the incremental margin multiplier scaled by the semantic distance ΔG_{ij} . This mechanism is the solution to the “static constraint” problem. Instead of treating all pairs uniformly, our margin now scales proportionally with the true semantic gap defined by the groups. This teaches GRFC that inter-group pairs with larger semantic gaps must have a larger predicted score difference than intra-group pairs with small gaps. The base ranking loss for the pair (i, j) is then:

$$\mathcal{L}_{\text{rank},ij} = \max(0, m_{ij} - (\hat{y}_i - \hat{y}_j)). \quad (7)$$

Applying Adaptive Error-Weighting: Having established a standard for separation (m_{ij}), we now introduce an adaptive strength for optimization (w_{ij}) using a dynamic re-weighting principle. First, we compute the raw prediction difference d_{ij} for a pair, representing the model’s absolute confidence in their relative order:

$$d_{ij} = \sigma(\hat{y}_i - \hat{y}_j). \quad (8)$$

Instead of applying a symmetric loss to all pairs, we compare this difference against the global batch distribution to identify which

pairs the model is struggling with. We calculate a relative confidence score C_{ij} by standardizing d_{ij} within the current batch:

$$C_{ij} = \frac{d_{ij} - \mu_d}{\sigma_d}, \quad (9)$$

where μ_d and σ_d are the mean and standard deviation of d_{ij} across all valid pairs in the batch.

A positive C_{ij} indicates an “easy pair” where the model’s confidence exceeds the batch average, while a negative C_{ij} signifies a “hard pair” (e.g., highly ambiguous or misordered samples). To filter out gradient noise from easy pairs, we transform this relative confidence metric into an adaptive, one-sided penalty weight w_{ij} :

$$w_{ij} = \text{ReLU}(-C_{ij}). \quad (10)$$

A uniform objective wastes optimization on easily distinguishable pairs. Our one-sided penalty acts as an adaptive filter, effectively up-weighting pairs with negative relative confidence to force the optimization process to focus exclusively on difficult pairs.

$$\mathcal{L}_{\text{group}} = \frac{1}{N_p} \sum_{s_i > s_j} w_{ij} \cdot \mathcal{L}_{\text{rank},ij}. \quad (11)$$

This allows GRFC to learn a smoother and more semantically coherent ordinal manifold.

A critical consideration is how this Adaptive Error-Weighting specifically benefits *multimodal* sentiment analysis, as opposed to being a generic unimodal optimization trick. In multimodal scenarios, the most challenging samples, which generate the highest prediction errors and thus receive the highest adaptive weights (w_{ij}), are predominantly instances characterized by cross-modal conflict or subtle, asynchronous non-verbal cues, such as a speaker smiling while expressing negative semantics.

Unlike standard point-wise regression that passively averages out these conflicting multimodal signals, our error-weighted Dynamic Margin acts as an implicit cross-modal conflict resolver. By actively forcing the optimization process to focus on these hard, highly ambiguous multimodal pairs, the unified encoder is compelled to learn nuanced cross-modal interactions rather than relying on a dominant modality like text. Consequently, while the mathematical formulation of GRFC is modality-agnostic, its dynamic hard-mining behavior is intrinsically tailored to untangle the complex, contradictory affective signals unique to multimodal data.

(b) *Distribution Regularization Loss* ($\mathcal{L}_{\text{reg}}$): To ensure the learned relative rankings do not “drift” in absolute terms, we regularize the predicted batch distribution ($\hat{y}$) to match the ground-truth batch distribution (s). This loss penalizes deviations in mean (M) and standard deviation (D) beyond a margin γ :

$$\mathcal{L}_{\text{reg}} = [(M_p - M_r)^2 - \gamma]_+ + [(D_p - D_r)^2 - \gamma]_+. \quad (12)$$

(c) *Boundary Loss* ($\mathcal{L}_{\text{bound}}$): Finally, to ensure predictions remain within the valid annotation range $[-S, S]$, we add a boundary loss:

$$\mathcal{L}_{\text{bound}} = \frac{1}{N} \sum_{i=1}^N \max(0, |\hat{y}_i| - S). \quad (13)$$

The total loss for Stage 1 is a weighted sum:

$$\mathcal{L}_{S1} = \lambda_1 \mathcal{L}_{\text{group}} + \lambda_2 \mathcal{L}_{\text{reg}} + \lambda_3 \mathcal{L}_{\text{bound}}. \quad (14)$$

3.3 Stage 2: Fine-Tuning

While Stage 1 establishes a robustly ordinal space, it inherently suffers from a “score offset” problem. Instead of employing a rigid point-wise constraint (e.g., Mean Absolute Error, MAE), Stage 1 relies exclusively on relative ordering ($\mathcal{L}_{\text{group}}$) and soft macroscopic alignment ($\mathcal{L}_{\text{reg}}$). Consequently, while the model successfully learns the correct semantic distances between samples, it lacks a definitive anchor for absolute score values (e.g., the exact neutral 0 point).

Although the scaled tanh function and Boundary Loss ($\mathcal{L}_{\text{bound}}$) prevent predictions from exceeding the valid annotation range, the entire learned ordinal manifold remains free to “drift” within these bounds. Therefore, Stage 2 primarily corrects this offset by introducing direct MAE-driven calibration, anchoring the learned relative manifold to exact absolute scores. To prevent catastrophic forgetting of the learned structure, we employ differential learning rates, applying a higher rate to the fusion/output layers and a lower rate to the foundational encoders. To preserve the learned structure, $\mathcal{L}_{\text{mae}}$ is integrated with the $\mathcal{L}_{\text{group}}$ and $\mathcal{L}_{\text{bound}}$ losses from Stage 1, which now maintain the ranking ability and act as regularizer:

$$\mathcal{L}_{S2} = \beta_1 \mathcal{L}_{\text{mae}} + \beta_2 \mathcal{L}_{\text{group}} + \beta_3 \mathcal{L}_{\text{bound}}. \quad (15)$$

The weight for MAE (β_1) is set higher, providing a strong gradient for calibration while others preserve the ordinal structure.

3.4 Extending to Binary Classification Tasks

GRFC is inherently suitable for regression because continuous labels provide a naturally dense space for evaluating relative prediction differences. However, adapting GRFC to binary classification presents distinct challenges. First, the task relies on sparse, discrete labels (e.g., Non-sarcastic/Sarcastic), which limits the fine-grained granularity required by our Adaptive Error-Weighting mechanism. Second, in binary classification, the existence of only two mutually exclusive classes invalidates the Dynamic Margin mechanism, which relies on multiple fine-grained, overlapping ordinal groups (e.g., G_0 to G_4) to calculate semantic distance.

Therefore, a principled adaptation was required. We decomposed the group-wise philosophy into a Separation Loss to push inter-class clusters apart and a Compactness Loss to pull intra-class samples together. The detailed methodology is provided in the Appendix.

4 Experiments

4.1 Implementation Details

We use DeBERTa-v3-base as the default text backbone; acoustic and visual features are aggregated via attention pooling and projected to a shared latent space before fusion. Training uses the AdamW optimizer with mixed precision, cosine learning-rate decay, gradient clipping, and distributed data parallelism. We conduct a two-part hyperparameter search via Optuna [1]. Stage 1 maximizes Pair-wise Accuracy (PairwiseAcc), while Stage 2 minimizes MAE.

As shown in Figure 3, in Stage 1 training, the Group-Aware Ranking Loss decreases steadily, and the Distribution Regularization Loss is active early in the process. Moreover, the Boundary Loss only activates during the later stages, which is a direct consequence of ordinal learning. As the ranking objective continuously pushes the pairs apart to establish clear relative distances, the overall score distribution naturally expands outward. Once the predictions hit

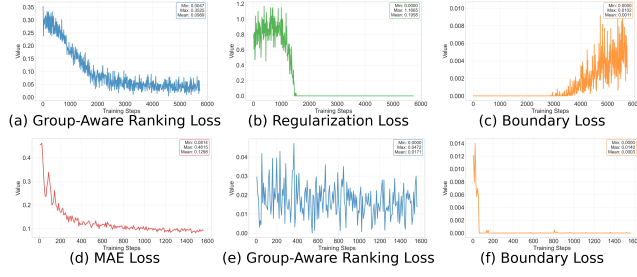


Figure 3: Training loss curves on CMU-MOSI dataset. (Top) Stage 1 loss. (Bottom) Stage 2 loss.

the boundaries, the Boundary Loss actively triggers to penalize and bound these values. In Stage 2 training, the primary MAE loss for calibration trends downward, the Group-Aware Ranking Loss remains stable, preserving the learned ranking, and the Boundary Loss is minimal, indicating few out-of-range predictions.

Notably, in Figure 3 (d), the training MAE converging to a minimum, which appears to contrast to the test MAE (0.581). Actually, this indicates that GRCF enters a benign interpolation regime. Because the optimal group structure established in Stage 1 robustly constrains the global ordinal manifold, Stage 2 interpolates these absolute scores locally via high-frequency adjustments without disrupting the global topology. Consequently, this prevents detrimental overfitting to intrinsic label noise and safely maps test samples to a robust semantic manifold.

4.2 Main Results on Sentiment Analysis

Table 1 presents the performance of GRCF on CMU-MOSI [47] and CMU-MOSEI [48] datasets. On the CMU-MOSI dataset, GRCF achieves SOTA results on all evaluation metrics. Conversely, on the CMU-MOSEI dataset, GRCF outperforms existing methods across all metrics, reaching SOTA levels consistently.

Table 2 shows the results on the Chinese-specific CH-SIMS v2 [20] dataset. To ensure a rigorous evaluation regarding backbone dependency and fair comparisons with prior works, we evaluate GRCF variants across different text encoders. While default DeBERTa-v3 already achieves highly competitive results, utilizing language-aligned encoders like mDeBERTa-v3 and BERT-chinese yields massive performance leaps. GRCF (BERT-chinese) establishes a completely new SOTA across all metrics, substantially outperforming the previous baselines (e.g., Acc5 58.4%, and Corr 0.744). This finding strongly proves two points: First, GRCF is robust and encoder-agnostic, capable of achieving peak performance under the BERT backbone setting used by earlier SOTA methods. Second, it demonstrates that GRCF inherently amplifies the representation power of the chosen encoder, rather than relying exclusively on DeBERTa’s capabilities.

4.3 Results on Humor and Sarcasm Detection

To evaluate the versatility of our framework, we adapt it for the binary classification tasks of Multimodal Sarcasm Detection on MUSTARD [3], and Multimodal Humor Detection on UR-FUNNY v2 [10]. It achieves an Acc2 of 75.00% on MUSTARD and 73.64% on

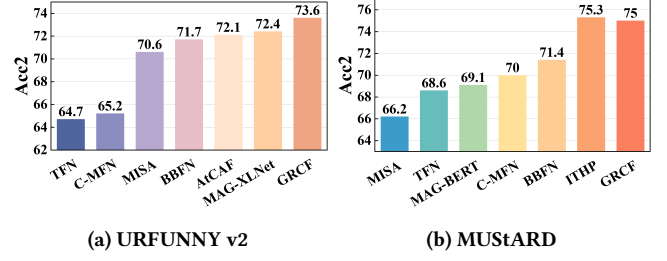


Figure 4: Results (Acc2) on the classification tasks.

UR-FUNNY v2, demonstrating that GRCF generalizes effectively, creating robust decision boundaries for complex classification tasks, not just for regression. Results are presented in Figure 4.

4.4 Ablation Experiment

To validate the necessity of Stage 1 components and the two-stage framework, we conducted an ablation study on CMU-MOSI, with results presented in Table 3.

(a) *Stage 1 Components*: Adaptive Error-Weighting and the Dynamic Margin are highly effective for datasets with granular sentiment labels like CMU-MOSI and CMU-MOSEI (range $[-3, 3]$). On CMU-MOSI, removing either Error-Weighting or the Dynamic Margin causes a substantial performance drop (e.g., -8.72% and -5.27% in Acc7, respectively), confirming their importance in establishing a well-separated ordinal manifold.

Interestingly, on CH-SIMS v2 (label range $[-1, 1]$), removing these components results in negligible changes or even a slight performance increase (e.g., w/o Error-Weighting improves Corr by +2.03%). We attribute this to the Signal-to-Noise Ratio (SNR) and the dataset’s intrinsic noise floor. The default grouping strategy ($K = 5$) aligns well with the continuous label distributions of CMU-MOSI and CMU-MOSEI, but clashes with the relatively discrete, coarse-grained annotations of CH-SIMS v2, whose annotations are relatively quantized into specific discrete steps. When the artificially defined Dynamic Margin attempts to impose fine-grained, continuous pair-wise constraints on these heavily quantized labels, GRCF is forced to overfit to this annotation discreteness rather than learning valid semantic differences. Consequently, the advantage of two components is less pronounced on such relatively discrete datasets, explaining the minimal performance drop upon their removal. This reveals that the granularity of ordinal grouping must be carefully calibrated not just to the label range, but to the intrinsic continuity and granularity of the dataset’s annotations.

(b) *Two-Stage Framework*: Removing Stage 1 and relying on direct regression causes a drop (e.g., -11.55% in Acc7 on CMU-MOSI), demonstrating the necessity of the ranking structural foundation. Moreover, removing Stage 2 results in a catastrophic performance collapse (e.g., -43.10% in Acc3 on CH-SIMS v2), proving that Stage 2 is indispensable for mapping ordinal space to absolute scores.

To further justify the necessity of the decoupled two-stage design, we compare our framework against a Single-Stage Bi-objective variant, which optimizes both $\mathcal{L}_{\text{group}}$ and $\mathcal{L}_{\text{mae}}$ concurrently (supplemented by $\mathcal{L}_{\text{bound}}$ and $\mathcal{L}_{\text{reg}}$). As demonstrated by the GRCF

Table 1: The comparison with baselines on CMU-MOSI and CMU-MOSEI. Acc2 and F1-score are calculated exclusively on positive and negative samples, with zero-valued instances excluded from the evaluation. The results labeled with [†] are obtained from original papers, and others are obtained from our experiments. Best results are in bold, and second-best are underlined.

Baseline Models	CMU-MOSI					CMU-MOSEI				
	Acc7 [†]	Acc2 [†]	F1-score [†]	MAE [↓]	Corr [†]	Acc7 [†]	Acc2 [†]	F1-score [†]	MAE [↓]	Corr [†]
Self-MM [44]	45.8	84.9	84.8	0.731	0.785	53.0	85.2	85.2	0.540	0.763
AtCAF [†] [14]	46.5	88.6	88.5	0.650	0.831	<u>55.9</u>	87.0	86.8	<u>0.508</u>	0.785
DLF [†] [39]	47.1	85.1	85.0	0.731	0.781	53.9	85.4	85.3	0.536	0.764
KuDA [7]	47.1	86.4	86.5	0.705	0.795	52.9	86.5	86.6	0.529	0.776
DEVA [†] [40]	46.3	86.3	86.3	0.730	0.787	52.3	86.1	86.2	0.541	0.769
C-MIB [24]	47.7	87.8	87.8	0.662	0.835	52.7	86.9	86.8	0.542	0.784
ITHP [41]	47.7	88.5	88.5	0.663	0.856	52.2	87.1	87.1	0.550	0.792
Multimodal Boosting [23]	<u>49.1</u>	88.5	88.4	0.634	0.855	54.0	86.5	86.5	0.523	0.779
DMD [17]	44.9	84.3	84.3	0.726	0.788	52.8	84.6	84.6	0.538	0.768
EMOE [6]	45.2	84.8	84.8	0.723	0.790	52.5	85.0	85.0	0.542	0.760
TMSO [†] [42]	47.4	87.2	87.2	0.687	0.809	55.6	86.4	86.2	0.526	0.766
MOAC [25]	48.6	<u>89.0</u>	<u>89.0</u>	<u>0.605</u>	<u>0.857</u>	54.3	<u>87.6</u>	<u>87.6</u>	0.512	<u>0.793</u>
TSDF [†] [26]	49.0	86.5	86.6	0.691	-	54.6	86.2	86.2	0.525	-
DecAlign [†] [30]	-	85.8	85.8	0.735	-	-	86.5	86.1	0.543	-
GRCF (Single-Stage Bi-objective)	45.3	88.1	88.1	0.650	0.849	56.0	86.6	86.5	0.516	0.792
GRCF (Two-Stage)	49.3	90.3	90.2	0.581	0.866	58.7	89.1	89.1	0.461	0.835

Table 2: The comparison with baselines on CH-SIMS v2.

Baseline Models	CH-SIMS v2					
	Acc5 [†]	Acc3 [†]	Acc2 [†]	F1-score [†]	MAE [↓]	Corr [†]
TFN [45]	53.3	70.9	78.1	78.1	0.322	0.662
LMF [21]	51.6	70.0	77.8	77.8	0.327	0.651
MFN [46]	55.4	72.7	79.4	79.4	0.301	0.712
Graph-MFN [48]	48.9	68.6	76.6	76.6	0.334	0.644
MISA [11]	47.5	68.9	78.2	78.3	0.342	0.671
MAG-BERT [31]	49.2	70.6	77.1	77.1	0.346	0.641
Self-MM [44]	53.5	72.7	78.7	78.6	0.315	0.691
MMIM [9]	50.5	70.4	77.8	77.8	0.339	0.641
AV-MC [20]	52.1	73.2	80.6	80.7	0.301	0.721
KuDA [7]	53.1	74.3	80.2	80.1	0.289	0.741
KAN-MCP[22]	<u>57.3</u>	<u>75.0</u>	81.6	81.7	<u>0.281</u>	<u>0.742</u>
GRCF (DeBERTa-v3)	54.4	74.3	81.7	81.6	0.305	0.691
GRCF (mDeBERTa-v3)	54.7	<u>75.0</u>	82.5	82.5	0.294	0.705
GRCF (BERT-chinese)	58.4	75.9	83.5	83.5	0.271	0.744

Table 3: Ablation study results on different datasets.

Method	CMU-MOSI		CMU-MOSEI		CH-SIMS v2	
	Acc7 [†]	Corr [†]	Acc7 [†]	Corr [†]	Acc3 [†]	Corr [†]
w/o Error-Weighting	45.0	0.835	53.9	0.736	74.4	0.705
w/o Dynamic Margin	46.7	0.849	53.9	0.749	74.3	0.695
w/o Stage 1	43.6	0.852	52.9	0.729	72.5	0.682
w/o Stage 2	37.7	0.855	42.5	0.695	42.3	0.634

(Single-Stage) results incorporated in Table 1, joint optimization yields significantly inferior performance across all metrics compared to our proposed two-stage approach. We attribute this to gradient conflicts: the relative ranking objective (which aggressively pushes boundary samples apart) and the absolute calibration objective (which anchors samples to specific numerical values) interfere with each other when optimized simultaneously.

Furthermore, decoupling improves computational efficiency. Table 8 shows that the two-stage model converges rapidly in merely 22 epochs (1816s). In contrast, the single-stage variant struggles to balance the conflicting objectives, requiring over 30 epochs (2971s) to reach a sub-optimal convergence. This proves that the two-stage paradigm is necessary to resolve optimization conflicts.

4.5 Qualitative Analysis

To provide an intuitive understanding of how proposed components contribute to predictions, we visualize the internal representation space via t-SNE for full model and four ablations in Figure 5. Instead of examining individual points, we highlight the case study samples whose ground-truth scores are smaller than -2.0 as red “X”s.

(a) *Full Model*: It enhances the structure established by Stage 1. The case study cluster in (a) is visibly more compact than the “ranking-only” cluster in (e). This shows that $\mathcal{L}_{\text{mae}}$ not only aligns the absolute scores but also acts as a fine-tuning mechanism, further improving the intra-class compactness of the learned manifold.

(b) *w/o Error-Weighting* & (c) *w/o Dynamic Margin*: These reveal why $\mathcal{L}_{\text{group}}$ is effective. Without Error-Weighting (b), the optimization is dominated by easy pairs while hard boundary samples are neglected, causing the continuous ordinal spectrum to fracture into disconnected extremes. Without the Dynamic Margin (c), the red “X” cluster scatters and “leaks” heavily into adjacent zones. This proves Error-Weighting ensures a smooth ranking transition by resolving ambiguous middle ground, while the Dynamic Margin enforces strict proportional boundaries.

(d) *w/o Stage 1*: It represents a complete structural failure. The case study samples are chaotically scattered across the entire representation space, indiscriminately mixing with “neutral” and even “positive” samples, demonstrating that $\mathcal{L}_{\text{mae}}$ is incapable of learning a semantically meaningful or ordinally-aware manifold.

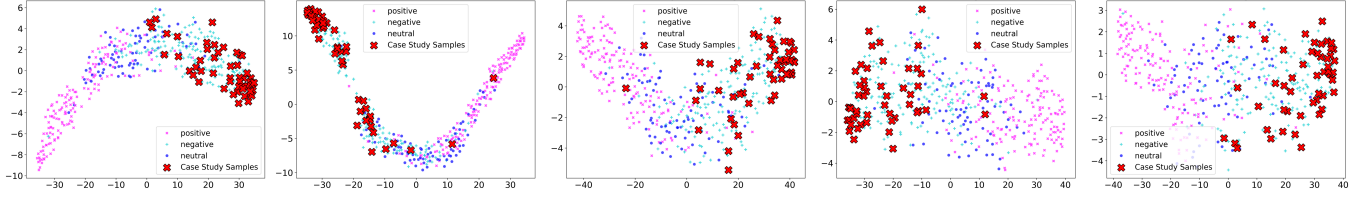


Figure 5: T-SNE visualization of ablations on CMU-MOSI. (a) Full model. (b) w/o Error-Weighting. (c) w/o Dynamic Margin. (d) w/o Stage 1. (e) w/o Stage 2.

Table 4: OLS regression analyzing predictors of the adaptive penalty weights on CMU-MOSI.

Variable	Coefficient	Std. Err.	t-statistic	P > t
Const	1.8443	0.007	278.811	< 0.001
Score Proximity	1.743e-08	7.43e-09	2.345	0.019
Conflict Count	0.0040	0.004	0.913	0.361

N = 3125 (top 5% hard pairs) | R-squared: 0.002

(e) w/o Stage 2: Stage 1 successfully establishes a clear ordinal gradient (placing positive samples on the left and negative on the right). However, the case study cluster remains loosely dispersed, proving Stage 1 builds the foundational ranking structure but requires Stage 2 for tight calibration.

In summary, our components are symbiotic. Stage 1 builds the foundational ordinal structure that direct regression completely misses, while Stage 2 refines this structure, leading to highly cohesive and strictly discriminative representations in the full model.

4.6 Analysis of Error-Weighting Mechanism

To understand how the Adaptive Error-Weighting mechanism focuses on “hard pairs”, we analyze the highest penalty weight samples from CMU-MOSI and MUSTARD. We hypothesize that the definition of “difficulty” is inherently task-dependent:

- For MSA, a modality congruence task, the challenge lies in discerning fine-grained degrees (e.g., score proximity).
- For MSD, a literal meaning subversion task, the challenge is to overcome strong unimodal biases (specifically, the literal meaning of text) via cross-modal contrast.

(a) On MSA: Difficulty is Driven by Fine-Grained Proximity: We conducted an Ordinary Least Squares (OLS) multiple regression analysis to distinguish whether the penalty weight (w_{ij}) on CMU-MOSI was driven by score proximity or superficial modality conflict.

As detailed in Table 4, when both factors were used as predictors, score proximity emerged as the only statistically significant factor ($p = 0.019$). Conflict count was not significant ($p = 0.361$), indicating it provides no independent explanatory power. Thus, in a congruent task like MSA, the true challenge is distinguishing fine-grained scores, and our adaptive weighting successfully identifies these semantically close pairs as hard cases.

(b) On MSD: Difficulty is Overcoming the Text Bias Trap: Conversely, sarcasm is defined by incongruity between literal text and true intent. To prevent the tautological bias of extracting unimodal

Table 5: OLS regression analyzing predictors of multimodal certainty on MUSTARD.

Variable	Coefficient	Std. Err.	t-statistic	P > t
Const	18.3717	2.207	8.325	< 0.001
Text Error Margin	1.1976	0.243	4.933	< 0.001

N = 539 | R-squared: 0.043 | Dep. Variable: Multimodal Certainty

predictions from a jointly trained network, we trained independent unimodal experts and defined the Text Error Margin, which means the degree to which the independent text model makes a confident, yet fundamentally incorrect, prediction based on literal semantics.

As shown in Table 5, an OLS regression reveals a highly significant positive relationship (coefficient = 1.1976, $p < 0.001$) between the Text Error Margin and the final multimodal model’s certainty. Strikingly, our analysis shows that when the text expert is completely misled by literal cues, the multimodal model exhibits 0.00% uncertainty. This counter-intuitive finding perfectly encapsulates the mechanics of sarcasm: the more heavily the literal text deviates from the true label, the stronger the cross-modal semantic contrast. Rather than falling into the “text bias trap,” the multimodal network leverages this stark contradiction to make highly certain predictions. Our Adaptive Error-Weighting mechanism facilitates this by dynamically assigning high penalties early in training to samples where textual cues are deeply misleading, forcing the network to abandon literal interpretations and learn deep, non-linear multimodal contrast.

Summary: These analyses validate that Error-Weighting mechanism is contextually adaptive and accurately captures the underlying nature of distinct tasks. In MSA, it targets fine-grained sentiment intensities. In MSD, it navigates the complex semantics of sarcasm by prioritizing samples that require overcoming deep literal text biases, proving that the model learns the true essence of cross-modal interactions rather than relying on naive heuristics.

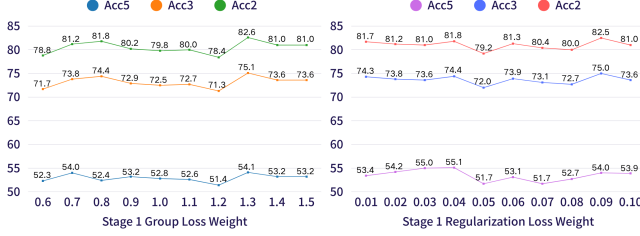
4.7 Analysis of Dynamic Margin Mechanism

To investigate how different grouping strategies for the Dynamic Margin affect performance, we conducted a comparative experiment on CH-SIMS v2 dataset (Table 6). We compare the “Overlap, 5” strategy (detailed in the Methodology 3) against three variants:

- “Overlap, 3” and “Overlap, 7”: Use 3 and 7 overlapping bins, respectively, to test granularity.
- “Strict, 5”: Uses 5 non-overlapping bins (e.g., [-1.0, -0.6), [-0.6, -0.2), ...) to test the importance of the overlap itself.

Table 6: Comparison of different Dynamic Margin strategies.

Strategy	Acc5 $\uparrow$	Acc3 $\uparrow$	Acc2 $\uparrow$	F1-score $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
Overlap, 3	54.6	73.8	81.2	81.2	0.307	0.689
Overlap, 5	<u>54.4</u>	74.3	81.7	81.6	0.305	0.691
Overlap, 7	54.1	<u>74.1</u>	<u>81.5</u>	81.5	<u>0.306</u>	0.692
Strict, 5	53.5	73.2	80.5	80.6	0.311	0.690

**Figure 6: Performance w.r.t the change of λ_1 and λ_2 .**

Results in Table 6 validate that “Overlap, 5” strategy significantly outperforms all other configurations in almost all metrics (e.g., 75.9 Acc2). The performance drop of the “Strict, 5” strategy (73.2 Acc2) confirms that modeling semantic ambiguity via overlapping intervals is critical. Furthermore, the degraded results from “Overlap, 3” and “Overlap, 7” suggest that 5 bins provide the optimal granularity for capturing the sentiment spectrum. Theoretically, increasing the number of overlapping groups (K) improves sentiment resolution, but this is limited by the intrinsic noise floor of the dataset (approximate to the SOTA MAE). If the grouping is finer than this noise level (e.g., 7 bins), the model overfits to annotation ambiguity rather than learning valid semantic differences. Therefore, $K = 5$ acts as an optimal trade-off constrained by semantic resolvability.

4.8 Hyperparameter Stability Experiments

In this section, we assess the impact of hyperparameters λ_1 and λ_2 , which denote the weights of $\mathcal{L}_{\text{group}}$ and $\mathcal{L}_{\text{reg}}$ in Stage 1, respectively, on CH-SIMS v2. As shown in Figure 6, GRCF delivers consistently satisfactory performance across a broad range of settings. For λ_1 , the metrics remain remarkably stable as the value varies from 0.6 to 1.5. Notably, when λ_1 is set to 1.3, the model achieves an Acc2 of 82.6, which surpasses the result of 81.7 reported in Table 2. Similarly, for λ_2 , performance fluctuations are marginal despite variations from 0.01 to 0.10. Interestingly, setting λ_2 to 0.04 yields a peak Acc5 of 55.1 and Acc3 of 74.4, again outperforming results in Table 2.

This phenomenon demonstrates that GRCF is highly robust to the specific magnitude of loss weights. The potential of GRCF can be further unlocked through hyperparameter tuning.

4.9 Robustness to Input Noise

To evaluate GRCF’s robustness against real-world sensor artifacts, we inject additive Gaussian input noise $N(0, \sigma^2)$ with varying standard deviations (σ) into the unimodal non-textual features. The results are presented in Table 7. Performance remains remarkably stable with minimal degradation, even at high noise levels. On

Table 7: Robustness to noise on non-textual modalities.

Noise	CMU-MOSI		CMU-MOSEI		CH-SIMS v2	
	MAE	Corr	MAE	Corr	MAE	Corr
0	0.581	0.866	0.461	0.835	0.305	0.691
0.01	0.581	0.866	0.461	0.835	0.305	0.691
0.05	0.581	0.866	0.461	0.835	0.306	0.691
0.1	0.582	0.866	0.461	0.835	0.305	0.691
0.15	0.581	0.865	0.461	0.835	0.305	0.692
0.2	0.583	0.866	0.461	0.835	0.305	0.692
0.3	0.577	0.867	0.461	0.835	0.305	0.691

Table 8: Computational cost on CMU-MOSI.

Model	Format	Inf. FLOPs (GFLOPs)	Params (m)
EMOE [6]	Point-wise	10.0	341.5
MOAC [25]	Pair-wise	10.2	186.9
GRCF	Pair-wise	7.7	186.6

CMU-MOSEI, the model is notably insensitive, showing no performance change up to $\sigma = 0.3$, suggesting a highly robust learned representation. Interestingly, we observe a slight regularization effect on CMU-MOSI, where the noise at $\sigma = 0.3$ marginally improves performance. This suggests that the noise may help the model disregard spurious, non-generalizable artifacts. In summary, the framework’s ability to withstand and, in some cases, leverage input noise demonstrates its high robustness.

4.10 Computational Cost and Scalability

While pair-wise ordinal learning introduces an $O(N^2)$ sample expansion during training, this computational overhead is strictly confined to the training phase. During inference, GRCF reverts to a standard point-wise forward pass, processing each test sample independently without complex multi-branch interactions. As shown in Table 8, due to our streamlined unified encoder, GRCF’s inference computational complexity (7.7 GFLOPs) is actually lower than these baselines. This reasonable one-time training cost yields a robust manifold for highly efficient, low-latency deployment.

5 Conclusion and Future Work

In this work, we introduced GRCF, which adapts the group-wise philosophy for ordinal representation learning in Affective Computing. Our core innovation is the Group-Aware Ranking Loss, which builds an ordinal-consistent foundation by focusing on hard-to-rank pairs via an Error-Weighting mechanism and enforcing semantically-proportional distances via a Dynamic Margin mechanism. GRCF achieves strong performance on regression benchmarks and demonstrates generalizability to classification tasks.

A key direction for future work is to address the current performance bottleneck in binary classification tasks. We attribute this limitation primarily to our existing multimodal fusion module; although it incorporates a gating mechanism, it remains relatively simple for modeling complex, heterogeneous cross-modal dynamics. Future research will focus on integrating more sophisticated fusion architectures to better capture nuanced affective cues and unresolved inter-modality conflicts.

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